João Marcos Amario de Sousa

Ph.D. Candidate in Genetics and Plant Breeding

Federal University of Viçosa (UFV)

Brazilian driver's license, category AD

Email · LinkedIn · GitHub repository · ORCID · Personal website

EDUCATION

2026 – present — **Visiting Scholar** · University of Illinois Urbana-Champaign, USA · Advisor: Prof. Alexander Lipka

2024 – present — **Ph.D. in Genetics and Plant Breeding** · Federal University of Viçosa (UFV) · Advisor: Prof. Kaio Olimpio

2022 – 2024 — **M.Sc. in Genetics and Plant Breeding** · Federal University of Viçosa (UFV) · Advisor: Prof. Maicon Nardino

2017 – 2020 — **B.Sc. in Agronomic Engineering** · Federal Institute of Tocantins (IFTO) · Advisor: Prof. Leonardo Correia

EXPERIENCE

Ph.D. in Genetics and Plant Breeding
Idata, Federal University of Viçosa · 2024 – present

- Model genotype-by-environment interaction with mixed models and integrate environmental covariates into genomic prediction to predict genotypes in untested environments.
- Lead a partnership project with GDM Seeds on continental-scale soybean data, combining phenotypic data and environmental covariates to support cultivar recommendation.
- Develop and maintain `qgtools`, an open-source collection of R functions for multi-environment trial and environmental covariate analyses, versioned on GitHub.
- Propose a pipeline to estimate the reliability and error of predictions for untested environments, showing breeding programs which predictions they can trust.

M.Sc. in Genetics and Plant Breeding
Federal University of Viçosa · 2022 – 2024

- Modeled genetic and residual covariances in tropical wheat populations to improve selection accuracy.

Agricultural Consultant
Fazenda Mundo Novo, Darcinópolis, TO, Brazil · 2022 – 2026

- Provided remote advisory on maize and pasture management based on soil, climate and performance data.

Farm Manager
Fazenda Mundo Novo, Darcinópolis, TO, Brazil · 2021 – 2022

- Managed maize and pasture production for livestock and led the field teams.

Technical Consultant
SENAR (CNA System), Imperatriz, MA, Brazil · 2020 – 2021

- Provided technical assistance to 30 beef cattle farms in Maranhão, focused on productivity and animal health.

Undergraduate Research, Sorghum Breeding
Embrapa Maize and Sorghum and IFTO · 2017 – 2020

- Evaluated dual-purpose biomass sorghum in the Embrapa Maize and Sorghum breeding program.

PUBLICATIONS

Modeling genotype-by-environment interaction for variety recommendation in tropical wheat
Sousa, J. M. A. et al. · *Crop Science* (2026) · View article ↗

Statistical genetics models with residual and genetic structures enhance the accuracy of selecting wheat populations
Sousa, J. M. A. et al. · *Euphytica* (2025) · View article ↗

Uncovering the parameters and genetic gain in selection strategies of wheat breeding populations
Sousa, J. M. A. et al. · *Revista Brasileira de Ciências Agrárias* (2025) · View article ↗

PRESENTATIONS

Aug 2026 — **Optimization of target population of environments for breeding programs** · XVII International Symposium on Genetics and Breeding (SIGM) · UFV, Viçosa, MG, Brazil

Nov 2025 — **Kernel-FA unlocks key environmental drivers of GEI** · XVI International Symposium on Genetics and Breeding (SIGM) · UFV, Viçosa, MG, Brazil

Nov 2024 — **Transforming tropical wheat recommendation: a fresh perspective with GIS-FA 1.1** · XV International Symposium on Genetics and Breeding (SIGM) · UFV, Viçosa, MG, Brazil

AWARDS

June 2026 — **1st place · UFV–Bayer Hackathon 2026** · Team Oraculum · Idata, UFV · Viçosa, MG, Brazil

July 2026 — **1st place · 3rd GDM Universities Challenge** · QGSL Laboratory, Idata, UFV · Cambé, PR, Brazil

Nov 2024 — **Honorable Mention · Professor Cosme Damião Award, XV SIGM** · Doctorate and Postdoctoral category · UFV, Viçosa, MG, Brazil

LEADERSHIP

2023 – 2024 — **Events Coordinator, GenMelhor** · Study Group in Genetics and Plant Breeding, UFV

2023 – 2024 — **Coordinator, InterGen** · Extension group bringing together seven Brazilian universities

PROFESSIONAL DEVELOPMENT

2026 – present — **Mentee, Bayer University Mentoring Program (B4U)** · One-on-one mentoring with a Bayer scientist

WORKING GENIUS

The Table Group's Working Genius assessment. My profile is **Loyal Finisher (ET | TE)**: I draw energy from seeing tasks through to completion and from supporting the people around me.

GENIUS
What energizes me
- **Tenacity**
- **Enablement**

COMPETENCY
I do it well, but it does not energize me
- **Invention**
- **Galvanizing**

FRUSTRATION
What drains me
- **Wonder**
- **Discernment**